

github-claude-mcp

Complete Setup & Usage Guide

v1.0.0 • github.com/mohit1jindal/github-claude-mcp

Part 1 — MCP Server (Claude → GitHub)



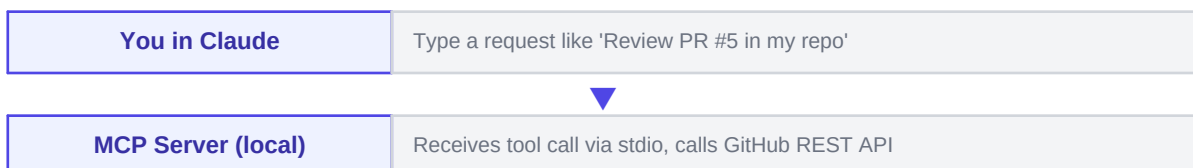
Runs locally on your machine. Gives Claude live read/write access to GitHub — repos, PRs, issues, commits, and file contents — directly from a conversation.

Part 2 — GitHub Action (GitHub → Claude)



Runs on GitHub's servers (free). Automatically fires Claude when a PR is opened or an issue is created. Posts AI reviews and triage comments back to GitHub.

Architecture





Built by Mohit Jindal • MIT License • 2026

Prerequisites

Everything you need before starting

- **Node.js 18+**
nodejs.org/en/download
Required to run the MCP server locally.
- **Git**
git-scm.com/downloads
Needed to clone the repository.
- **Claude Code CLI or Claude Desktop**
claude.ai/code or claude.ai/download
The AI client that the MCP server connects to.
- **GitHub Account**
github.com
Required for both parts.
- **Anthropic Account**
console.anthropic.com
Required only for the GitHub Action (Part 2).

Part 1 — MCP Server Setup

Gives Claude live GitHub access from any conversation

1 Clone and build the project

```
git clone https://github.com/mohit1jindal/github-claude-mcp.git
cd github-claude-mcp
npm install
npm run build

# You should see: github-claude-mcp running on stdio
```

2 Create a GitHub Personal Access Token

- Go to github.com/settings/tokens
- Click **Generate new token (classic)**
- Give it a name, e.g. *claude-mcp*
- Set expiration — 90 days recommended
- Select scopes: **repo** (full repo access) and **read:org**
- Click **Generate token**
- **Copy the token immediately** — it starts with `ghp_...` and you won't see it again

3 Configure Claude Code (or Claude Desktop — see below)

Edit `~/.claude/settings.json` (Windows: `C:\Users\YourName\.claude\settings.json`):

```
{
  "mcpServers": {
    "github": {
      "command": "node",
      "args": ["C:\\path\\to\\github-claude-mcp\\dist\\index.js"],
      "env": {
        "GITHUB_TOKEN": "ghp_your_token_here"
      }
    }
  }
}
```

■ ■ Windows: use double backslashes `\\` or forward slashes `/` in the path.

Claude Desktop alternative

Edit `%APPDATA%\Claude\claude_desktop_config.json` (Windows) or `~/Library/Application Support/Claude/claude_desktop_config.json` (Mac) with the same `mcpServers` block above.

Restart Claude Desktop after saving.



Verify the connection

Open Claude Code and type:

```
"List my GitHub repos"
```

■ Claude calls `list_repos` and returns your repositories within seconds.

Troubleshooting

■ **GITHUB_TOKEN environment variable is required**

→ The env var isn't reaching the server. Double-check the "env" key in settings.json.

■ **Module not found / path error**

→ Run `npm run build` first. Make sure you're using the absolute path to `dist/index.js`.

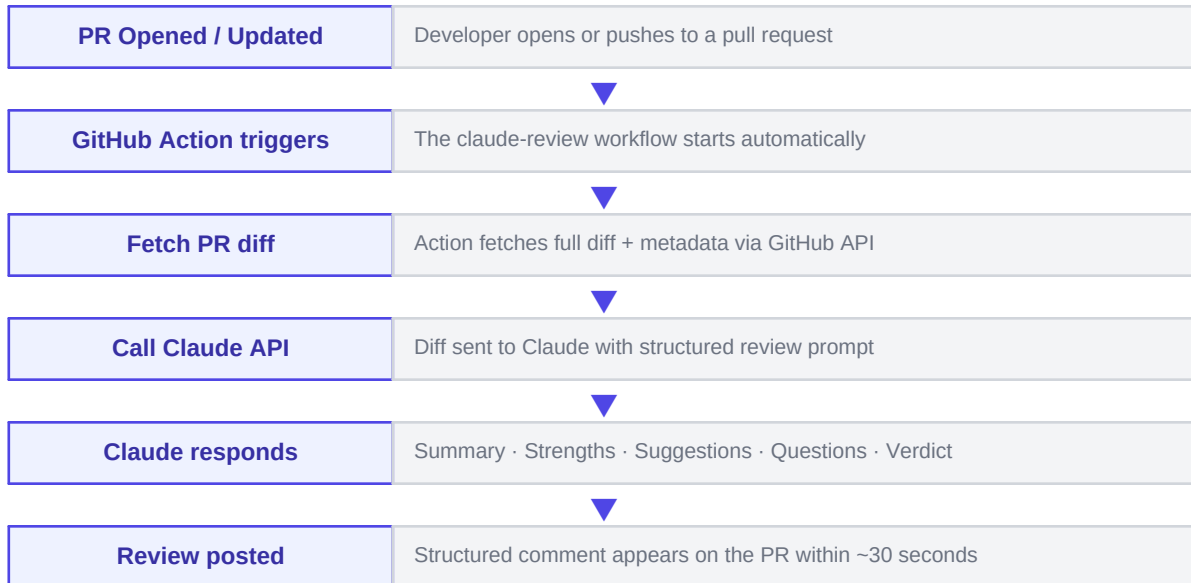
■ **No tools appearing in Claude**

→ Restart Claude Code/Desktop after editing settings.json. Check for JSON syntax errors (no trailing commas).

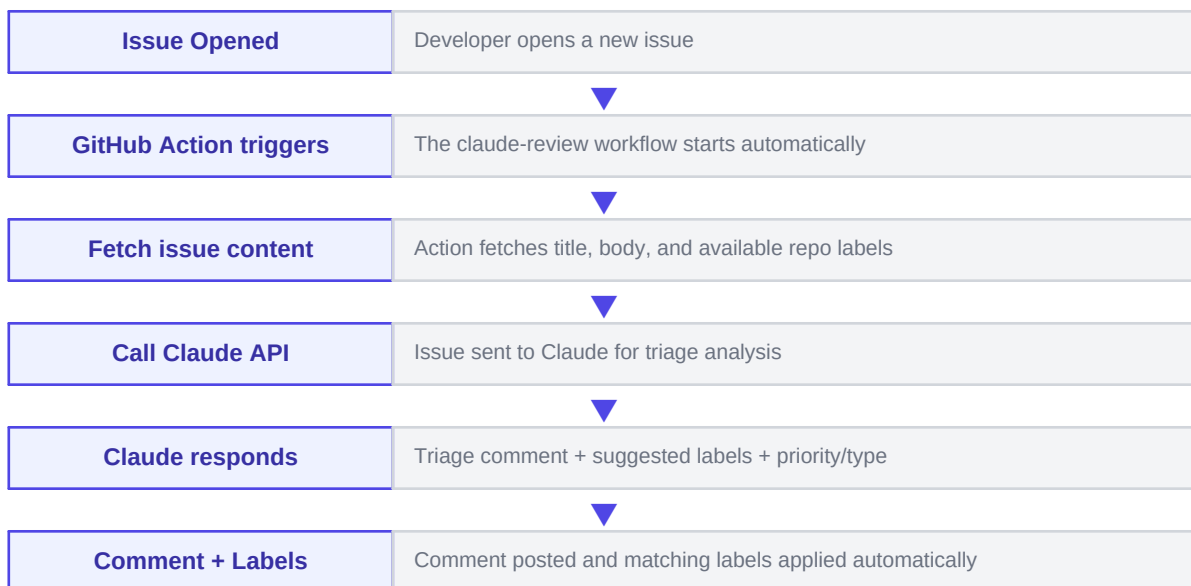
Part 2 — GitHub Action Setup

Automated Claude reviews on every PR and issue

PR Review — How it works



Issue Triage — How it works



1 Get an Anthropic API Key

- Go to **console.anthropic.com**
- Sign in or create a free account
- Click **API Keys** in the left sidebar → **Create Key**

- Name it (e.g. *github-claude-mcp*)
- Copy the key — starts with `sk-ant-`...

Event	Approx tokens	Approx cost
Small PR review	~1,000–2,000	~\$0.005
Large PR review	~3,000–4,000	~\$0.015
Issue triage	~500–1,000	~\$0.002

■ For a personal repo with low traffic: typically under \$1/month.

2

Add ANTHROPIC_API_KEY as a GitHub Secret

- Open the repository on GitHub
- Click **Settings** (top navigation bar of the repo)
- Left sidebar: **Secrets and variables** → **Actions**
- Click **New repository secret**
- Name: **ANTHROPIC_API_KEY** ← must be exactly this
- Value: paste your `sk-ant-...` key
- Click **Add secret**

■ GITHUB_TOKEN is automatically injected by GitHub into every workflow run — you do NOT need to create it.

3

Add the workflow file

Create `.github/workflows/claude-review.yml` in your repository:

```
name: Claude AI Review

on:
  pull_request:
    types: [opened, synchronize] # New PRs + new commits to existing PRs
  issues:
    types: [opened] # New issues only

permissions:
  pull-requests: write # Post PR review comments
  issues: write # Post issue comments + apply labels
  contents: read # Read repo code

jobs:
  claude-review:
    runs-on: ubuntu-latest
    if: ${{ github.event_name == 'issues' || !github.event.pull_request.draft }}
```

```

steps:
  - name: Claude AI Review & Triage
    uses: mohit1jindal/github-claude-mcp@v1
    with:
      anthropic_api_key: ${ secrets.ANTHROPIC_API_KEY }
      github_token: ${ secrets.GITHUB_TOKEN }
      model: "claude-sonnet-4-6"    # Optional
      max_tokens: "2048"          # Optional

```

4 Test it

- Commit and push the workflow file to your repo's **default branch** (main/master)
- Open a new Pull Request in the repo
- Check the **Actions** tab — you'll see *Claude AI Review* running
- A structured review comment will appear on the PR within ~30 seconds

What the PR review looks like

```

## Summary
This PR adds input validation to the invoice processing pipeline.

## Strengths
- Clean separation between validation and processing logic
- Error messages are specific and actionable

## Suggestions
- InvoiceValidator.java:45 – extract hour threshold to a named constant
- validateLineItems() is 80+ lines – consider splitting into smaller methods
- Add a test for lineItem.getHours() == null edge case

## Questions
- Should zero-hour expense line items be allowed through?

## Verdict
COMMENT – logic is sound, suggestions are minor improvements

--- Review by github-claude-mcp using claude-sonnet-4-6

```

Customization Options

PRs only (skip issue triage)	Remove the issues: block from the on: section
Issues only (skip PR review)	Remove the pull_request: block from the on: section
Deeper reviews (Opus model)	Set model: "claude-opus-4-7" in the with: block
Longer reviews	Set max_tokens: "4096" in the with: block

Review draft PRs too	Remove the if: condition from the job
All repos in an org	Add ANTHROPIC_API_KEY as an Organization Secret

Troubleshooting

Symptom	Cause	Fix
■ Action doesn't trigger	Workflow file not on default branch	→ Push <code>claude-review.yml</code> to main or master
■ Resource not accessible by integration	Missing permissions block	→ Add the <code>permissions:</code> section exactly as shown above
■ Bad credentials / auth error	Wrong secret name	→ Check Settings → Secrets — must be exactly <code>ANTHROPIC_API_KEY</code>
■ Action runs but no comment appears	Token lacks write permission	→ Ensure <code>pull-requests: write</code> and <code>issues: write</code> are in <code>permissions:</code>
■ Draft PRs not reviewed	By design — drafts are skipped	→ Remove the if: condition from the job to review drafts

MCP Server — Tools Reference

Everything you can ask Claude to do with your GitHub

Tool	Description	Example Prompt
list_repos	List repos for a user or org with language, stars, issue count	"List all my public GitHub repos"
get_repo	Detailed repo info: languages, topics, contributors, stats	"Tell me about mohit1jindal/github-claude-mcp"
get_file_contents	Read any file from any branch or commit SHA	"Read src/index.ts from my repo on main branch"
list_pull_requests	List PRs with state, labels, authors, draft status	"Show me all open PRs in owner/repo"
get_pull_request	Full PR details including unified diff	"Get the full diff for PR #5 in owner/repo"
post_pr_review	Post a review: APPROVE, REQUEST_CHANGES, or COMMENT	"Post a review on PR #5: LGTM, good error handling"
list_issues	List issues with label and state filters	"List all open bug issues in owner/repo"
create_issue	Create a new issue with labels and assignees	"Create an issue: Fix null pointer in validator [bug]"
add_issue_comment	Add a comment to any issue or PR number	"Add a comment to issue #12: Fixed in PR #15"
list_commits	Recent commits for a repo or specific file path	"Show the last 10 commits on the main branch"
get_commit	Full commit details with per-file diffs	"What changed in commit abc1234?"

Quick Reference

Part 1 — MCP Server

1. git clone <https://github.com/mohit1jindal/github-claude-mcp>.git
2. cd github-claude-mcp && npm install && npm run build
3. Create GitHub token at github.com/settings/tokens (scopes: repo, read:org)
4. Add to ~/.claude/settings.json with "command": "node", "args": [...dist/index.js]
5. Ask Claude: "List my GitHub repos" to verify

Part 2 — GitHub Action

1. Get Anthropic API key from console.anthropic.com
2. Add ANTHROPIC_API_KEY secret in repo Settings → Secrets → Actions
3. Create `.github/workflows/claude-review.yml` with the workflow content
4. Commit and push to the default branch
5. Open a PR to trigger the first review